

CARLOS ARIEL ORTEGA GUERRÓN

Data Scientist | Biomedical Engineer

Quito, Ecuador | +593 995517709 | arielguerron@gmail.com

LinkedIn: linkedin.com/in/ariel-ortega-guerron/ | GitHub: github.com/ArielOGc1

PROFESSIONAL SUMMARY

Data Scientist with training in Biomedical Engineering, specialized in statistical analysis, machine learning, and data visualization. Technical experience in clinical environments with demonstrated ability to develop analytical solutions using Python and SQL. Competencies in regression, classification, data preprocessing, and predictive modeling. Practical training in biomedical systems and medical equipment with focus on technical precision and quality standards.

TECHNICAL SKILLS

Python | SQL | Pandas | NumPy | Scikit-learn | Matplotlib | Seaborn | Plotly | Streamlit | TensorFlow | Keras | GitHub | Git | Bash | Linux | Excel | Power BI | Machine Learning | Statistical Analysis | Predictive Modeling | Linear Regression | Classification | Bootstrapping | Data Preprocessing | Data Visualization | Biomedical Systems | Dental Equipment | Clinical Protocols

PROFESSIONAL EXPERIENCE

Technical Assistant – Dental Medical Solutions

Internship | September 2024 – October 2025

- Assisted in technical maintenance of critical dental equipment (compressors, suction systems, dental chairs, turbines) under supervision of specialized engineer.
- Developed practical experience in technical diagnosis, preventive maintenance procedures, and safety protocols in clinical environments.
- Acquired competencies in precision technical work, biomedical equipment operation, and clinical quality standards compliance.
- Performed technical support tasks contributing to continuous operability of medical equipment in clinical care environments.

Technical Volunteer – Biomedicina Masters

Volunteer | February 2024 – September 2025

- Supported logistics coordination and organization of biomedical equipment training programs for healthcare professionals.
- Assisted in technical training and demonstrations of medical equipment, contributing to knowledge transfer in biomedical systems.
- Managed inventory of medical equipment and devices, documenting material movement and ensuring resource availability for training programs.

PROJECTS

Credit Scoring System – Loan Risk Classification

2026

Developed machine learning predictive model for assessing loan default probability with interactive web interface.

Technologies: Python, Pandas, Scikit-learn, Streamlit, Matplotlib, NumPy | Methodology: Financial data preprocessing, outlier treatment, resampling techniques for class imbalance, validation with F1-Score and ROC-AUC.

Oil Well Location Optimization through Regression Analysis

2025

Developed regression model to optimize new well placement based on reserve volume and return on investment.

Technologies: Python, Scikit-learn, Pandas, Bootstrapping | Methodology: Exploratory analysis, multivariate linear regression with risk evaluation through Bootstrapping.

Telecom Plan Classification Model using Machine Learning

2025

Built classification model for user behavior analysis and optimal telecom plan recommendations.

Technologies: Python, Scikit-learn, Pandas, Matplotlib | Methodology: Database consolidation, feature engineering, model training (Decision Trees, Random Forest).

EDUCATION

Biomedical Engineering – Pontifical Salesiana University (Graduate - July 2026)

Data Scientist – TripleTen (Online Bootcamp, 2025)

Python Programming Specialization – Udemy (2025)

LANGUAGES

Spanish (Native) | English (B2)